Jiaheng Chen

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Education

University of Chicago Chicago, US

- Ph.D. in Computational and Applied Mathematics 2022 – present
- Advisor: Daniel Sanz-Alonso

Shanghai Jiao Tong University Shanghai, China

- B.S. in Mathematics and Applied Mathematics, Zhiyuan College 2018 – 2022

Research Interests

Applied mathematics, statistics, data science, scientific machine learning, operator learning, high-dimensional probability, uncertainty quantification, inverse problems, and Monte Carlo methods.

Publications

[1] Optimal estimation of structured covariance operators. With O. Al-Ghattas, D. Sanz-Alonso and N. Waniorek.
SIAM/ASA Journal on Uncertainty Quantification, to appear, (2026+).

[2] High-dimensional quasi-Monte Carlo via combinatorial discrepancy. With H. Jiang and N. Kirk.
Journal of Complexity, 96, 102053, 1-23, (2026).

[3] Sharp concentration of simple random tensors II: asymmetry. With D. Sanz-Alonso.
Information and Inference, 15(2), 1-46, (2026).

[4] On the estimation of Gaussian moment tensors. With O. Al-Ghattas and D. Sanz-Alonso.
Electronic Communications in Probability, 30, 1-15, (2025).

[5] Sharp concentration of simple random tensors. With O. Al-Ghattas and D. Sanz-Alonso.
Information and Inference, 14(4), 1-41, (2025).

[6] Precision and Cholesky factor estimation for Gaussian processes. With D. Sanz-Alonso.
SIAM/ASA Journal on Uncertainty Quantification, 13(3), 1085-1115, (2025).

[7] Covariance operator estimation: sparsity, lengthscale, and ensemble Kalman filters. With O. Al-Ghattas, D. Sanz-Alonso and N. Waniorek.
Bernoulli, 31(3), 2377-2402, (2025).

[8] A machine learning framework for geodesics under spherical Wasserstein-Fisher-Rao metric and its application for weighted sample generation. With Y. Jing, L. Li and J. Lu.
Journal of Scientific Computing, 98(5), 1-34, (2024).

Preprints

[1] Adaptive Bayesian learning of linear operators. With W. Li and D. Sanz-Alonso.
In preparation.

[2] Minimax learning of Fourier integral operators. With D. Sanz-Alonso.
In preparation.

[3] Sharp Frobenius-norm concentration for sample moment tensors. With D. Sanz-Alonso.
arXiv: 2608.11084

[4] Optimal multiscale learning of linear operators. With D. Sanz-Alonso.
arXiv: 2606.16913

[5] Concentration inequalities for sample cross-covariances. With D. Sanz-Alonso.
arXiv: 2605.16733

[6] Convergence rates for learning pseudo-differential operators. With D. Sanz-Alonso.
arXiv: 2601.04473

[7] Fluctuation suppression and enhancement in interacting particle systems. With L. Li.
arXiv: 2204.07757

Fellowships and Awards

- William Rainey Harper Dissertation Fellowship, University of Chicago 2026
  Awarded in recognition of a PhD candidate's record of achievement and professional promise; one of the University of Chicago's highest honors
- Travel Award, SIAM Conference on Uncertainty Quantification 2026
- Neubauer Family Distinguished Doctoral Fellowship, University of Chicago 2022 – 2027
- Zhiyuan Outstanding Student Scholarship, Shanghai Jiao Tong University 2022
- Excellent Undergraduate Thesis, Shanghai Jiao Tong University 2022
- Shanghai Outstanding Graduate 2022
- Outstanding Winner, SIAM Award, COMAP Scholarship Award 2021
  Mathematical Contest in Modeling and Interdisciplinary Contest in Modeling
- National Scholarship, Ministry of Education of China 2020
- Zhiyuan Honors Scholarship, Shanghai Jiao Tong University 2018 – 2022

Conferences and Seminars

- AMS Spring Central Sectional Meeting, Iowa City, US Apr 17 – 18, 2027
  Session: "Tensors, Transport, and Particle Dynamics in High-Dimensional Sampling and Inference"
- SIAM Conference on Computational Science and Engineering, Pittsburgh, US Feb 22 – 26, 2027
  Minisymposium: "Probability Models and Distributional Data in Scientific Machine Learning"
- Joint Mathematics Meetings (JMM), Chicago, US Jan 12 – 15, 2027
  SIAM Minisymposium on Mathematical Foundations of Operator Learning
- SIAM Conference on Mathematics of Data Science, Salt Lake City, US Nov 16 – 20, 2026
  Minisymposium: "Recent Advances in Operator Learning: Theory, Methods, and Applications"
- SIAM Annual Meeting, Cleveland, US Jul 6 – 10, 2026
  Minisymposium: "From Theory to Computation: Modern Methods for PDE Inverse Problems"
- SIAM Conference on Uncertainty Quantification, Minneapolis, US Mar 22 – 25, 2026
  Minisymposium: "Recent Advancements in Operator Estimation and Learning"

- CAM & Statistics Student Seminar, University of Chicago — Mar 4, 2026
- Joint Statistical Meetings (JSM), Nashville, US — Aug 2 – 7, 2025
  Section: "Functional and Shape-Constrained Data Analysis: Methods and Applications"
- DACO Seminar, Department of Mathematics, ETH Zurich (*virtual*) — May 22, 2025
- Stat & CAM Student Research Poster Day, University of Chicago — May 20, 2025
- Workshop "Statistics Meets Tensors: Methodology, Theory, and Applications", Institute for Mathematical and Statistical Innovation (IMSI) — May 5 – 9, 2025
- Reading Seminar on Stochastic Many-body Systems (*virtual*) — Apr 2, 2025
- Workshop "Kernel Methods in Uncertainty Quantification and Experimental Design", Institute for Mathematical and Statistical Innovation (IMSI) — Mar 31 – Apr 4, 2025
- Institute of Natural Sciences, Shanghai Jiao Tong University — Sep 1, 2024
- SIAM Conference on Uncertainty Quantification, Trieste, Italy — Feb 27 – Mar 1, 2024
  Minisymposium: "Statistical and Data-Assisted Modeling Approaches for Forecasting and Uncertainty Quantification of Complex Multiscale Systems in Real-World Applications"
- CAM & Statistics Student Seminar, University of Chicago — Oct 10, 2023

Conference Organization

- SIAM Conference on Computational Science and Engineering, Pittsburgh, US — Feb 22 – 26, 2027
  Co-organizer, minisymposium "Error Analysis for Operator Learning", with Margaret Trautner
- SIAM Conference on Uncertainty Quantification, Minneapolis, US — Mar 22 – 25, 2026
  Co-organizer, minisymposium "Recent Advancements in Operator Estimation and Learning", with Omar Al-Ghattas and Nathan Waniorek

Teaching

Teaching Assistant at University of Chicago

- STAT/CAAM 31511: Monte Carlo Simulation — Spring 2026
- STAT/CAAM 31521: Applied Stochastic Processes — Spring 2025
- STAT/CAAM 31050: Applied Approximation Theory — Spring 2024
- STAT/CAAM 38100: Measure-Theoretic Probability I — Winter 2024
- STAT/CAAM 31150: Inverse Problems and Data Assimilation — Autumn 2023

Referee Service

Advances in Computational Mathematics, Biostatistics